Heterogeneity-Aware Adaptive Federated Distillation for Smart Campus Energy Management

Research Progress Update

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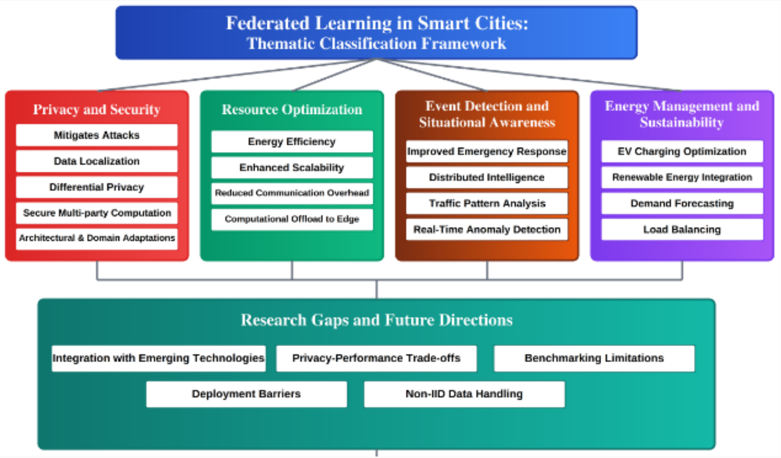
Edge ML & Smart City/Campus



Alterkawi and Dib (2025), Figure 2: Challenge-oriented taxonomy of FL applications in smart cities.

Energy management and resource optimization are primary FL challenges in urban environments.

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Domain	Data Sources	Characteristics	Challenges for FL
Transportation	Traffic sensors, GPS traces, vehicular networks	Spatio-temporal, high frequency, multimodal	Non-IID, real-time latency constraints
Energy	Smart meters, solar inverters, EV chargers	Periodic, household-level, sensitive usage data	Privacy, seasonal variation, adversarial risks
Healthcare	Wearables, hospital EHRs, emergency services	Multimodal, irregular sampling, highly sensitive	Regulatory compliance, heterogeneity
Public Safety	Surveillance feeds, incident reports, police IoT	Video, text, event-driven data	Large-scale streaming, privacy, security risks
Environment	Weather stations, pollution sensors, waste systems	Continuous, geographically distributed	Missing values, spatial heterogeneity

Alterkawi and Dib (2025), Table 2: Representative smart city data sources and characteristics.

Urban energy and environmental data are multimodal, continuous, and heterogeneous.

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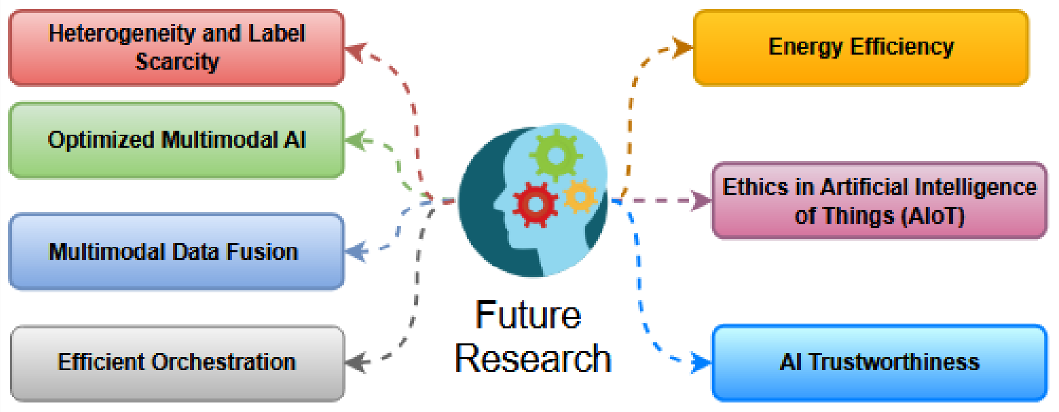
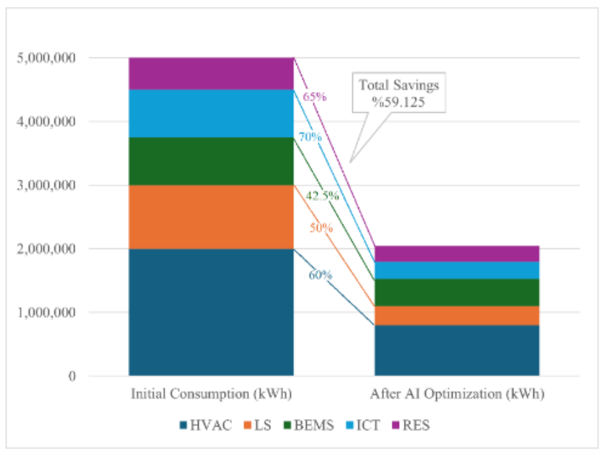


Cajas Ordóñez et al. (2025), Figure 8: Future research areas for machine learning at the edge.

Data heterogeneity, energy efficiency, and orchestration are the main bottlenecks for Edge ML.

Smart Campus Energy Management



Manassra and Işık (2025), Figure 8: Theoretical energy savings in a mid-sized smart campus.

AI optimization can achieve up to 60% energy savings for campus HVAC systems.

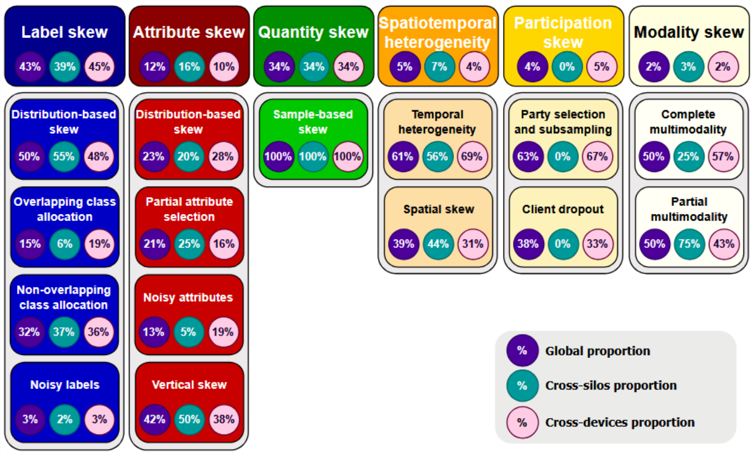
Smart Campus Energy Management

			AI Technology			Effect	
System	Reference Study	Data Type	Learning	Optimization	Control	Energy Saving	Other Benefit
HVAC	(Purdon et al., 2013)	Text Data	Occupant feedback, Classifier	-	MFPC	60%	-
RES	(Mayer et al., 2016)	Time series data, Text data	White box model, FNN	MLP	MPC	50%	Cost saving 50%
BEMS	(Salakij et al., 2016)	Time series data, Text data	Building model, MLP		MPC	42.60%	Prediction MAPE up to 6.3%
LS	(Kolokotsa, 2003)	Time series data	RNN	-	Adaptive MPC	70%	-
ICT	(Sinclair et al., 2013)	Map/image data	CNN	-	Handover management	65%	-

Manassra and Işık (2025), Table 2: Artificial intelligence technologies in smart campus systems.

Different ML models (such as MLPs, CNNs, and RNNs) target specific campus subsystems.

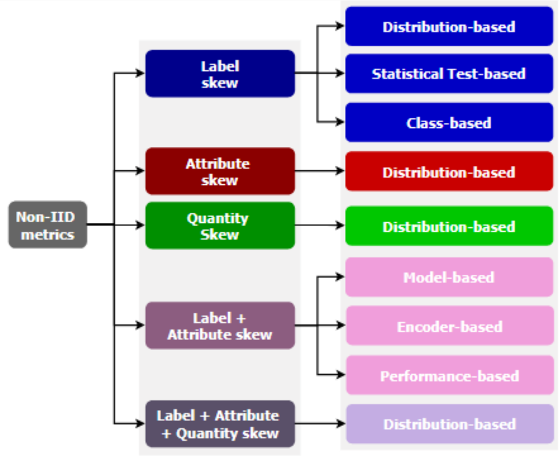
Non-IID Quantification & Compression



Jimenez et al. (2024), Figure 7: Non-IID skew and distribution visualization.

Non-IID data skew alters data distributions across federated edge nodes.

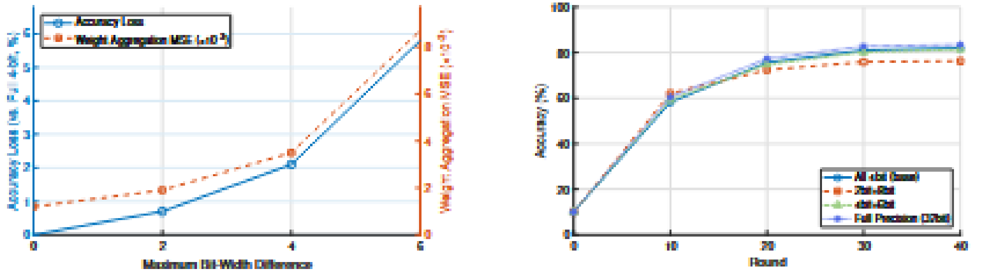
Non-IID Quantification & Compression



Jimenez et al. (2024), Figure 10: Taxonomy for non-IID metrics in Federated Learning.

Distribution-based metrics, like the Wasserstein distance, are required to measure attribute skew in continuous time-series energy data.

DynFed & Dynamic Quantization



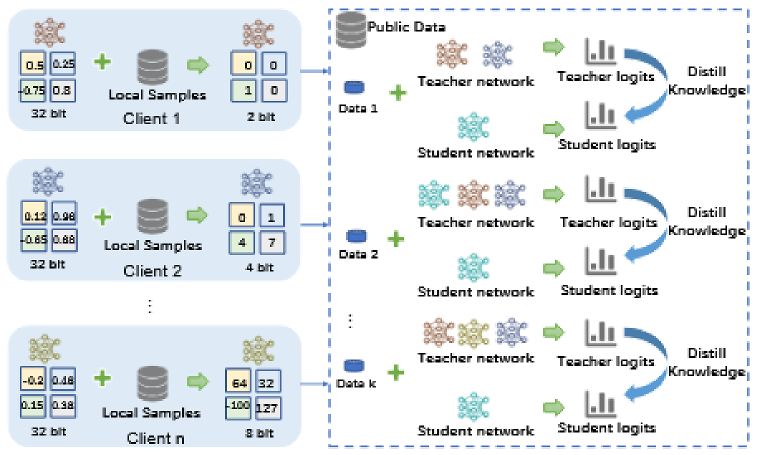
(a) Impact of Extreme Bit-Width Disparity on Model Convergence

(b) Nonlinear Aggregation Error Induced by Bit-Width Disparity

He et al. (2025 - DynFed), Figure 1: Effects of heterogeneous quantization bit-widths.

Bit-width disparities degrade standard FL aggregation, requiring a dynamic approach.

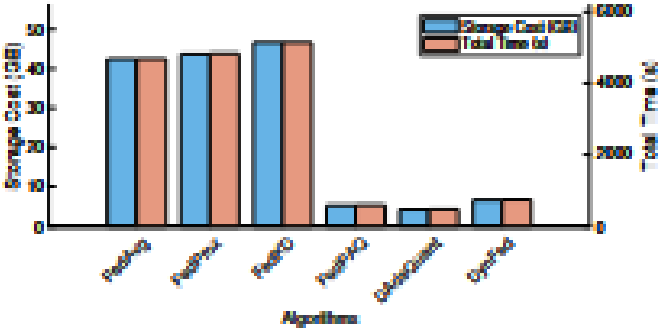
DynFed & Dynamic Quantization



He et al. (2025 - DynFed), Figure 2: Overview of the training procedure proposed by DynFed.

The architecture relies on clients with varying bit-widths (e.g., 2-bit, 4-bit, 32-bit) extracting and distilling logits to a central server.

DynFed & Dynamic Quantization

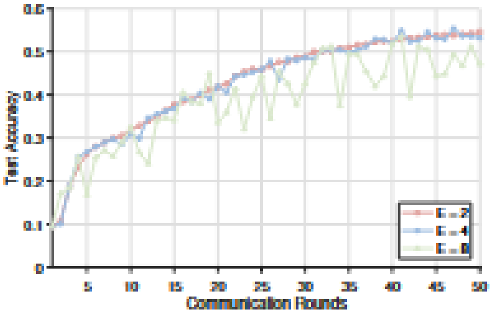


(a) Storage and time cost

He et al. (2025 - DynFed), Figure 3a: Storage Cost and Total Time comparison.

Dynamic quantization reduces communication overhead compared to uncompressed FedAvg or FedProx.

DynFed & Dynamic Quantization

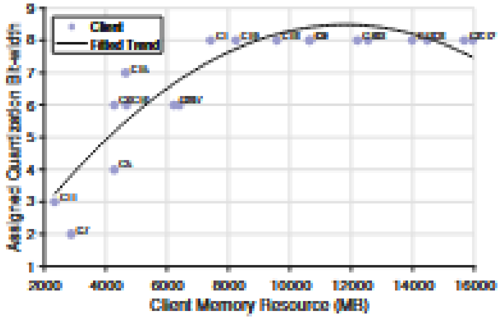


(b) Accuracy vs. client heterogeneity

He et al. (2025 - DynFed), Figure 3b: Test Accuracy under varying levels of client heterogeneity.

The algorithm maintains accuracy even when a large proportion of clients face resource limits.

DynFed & Dynamic Quantization



(c) Adaptive quantization by memory

He et al. (2025 - DynFed), Figure 3c: Adaptive quantization bit-width allocation mapped to client memory.

Assigned quantization bit-widths must positively correlate with a client's hardware capacity.

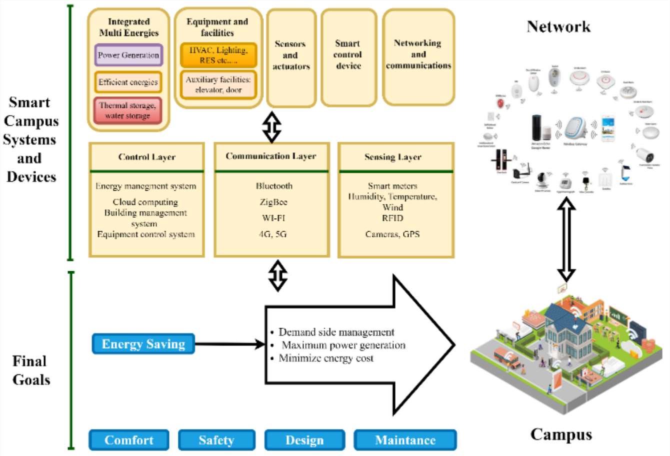
DynFed & Dynamic Quantization

RQ	AQ	ATS	Acc.	Overhead	Totel Time
×	×	×	81.87	20.625 G	738 s
✓	×	×	78.71	5.228 G	517 s
✓	✓	×	79.83	4.028 G	554 s
✓	✓	✓	82.39	5.656 G	738 s

He et al. (2025 - DynFed), Table 2: Ablation study of major components.

Combining Resource-Aware Quantization, Adaptive Loss Dynamics, and Active Teacher Selection yields peak accuracy.

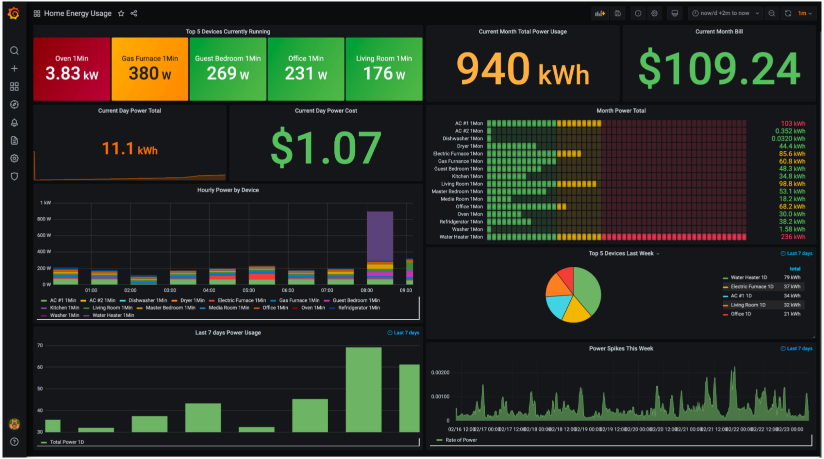
Real-World Deployment



Manassra and Işık (2025), Figure 9: Architecture of the AI-driven smart campus energy management system.

Real-world campus deployments may require a physical topology consisting of Sensing, Communication, and Control layers.

Real-World Deployment



Grafana Smart Energy Dashboard.

A potential practical extension of my research.

Commitments for This Week

- Continue studying and analyzing state-of-the-art adaptive quantization FL algorithms/protocols (e.g., DynFed, DAdaQuant).
- Continue investigating real-world pilot tests and operational implementations of FL for smart-city/campus energy management.
- Discuss with Sir Fritz the specifics of deploying IoT devices, sensors, etc., in DLSU and ALTDSI.

Thank You!

Questions & Answers